

Table 1: Kuo Quadrupole R0500 (fitting region: cube ± 70 μm , full FEM mesh)

$$\text{Error} = \frac{1}{N} \sum_{i=1}^N \frac{|\mathbf{B}_{model} - \mathbf{B}_{FEM}|_i}{|\mathbf{B}_{FEM}|_i} \cdot 100$$

Kuo Quadrupole (Error=7.12%)		
Pole	$ \ell $ [μm]	Tip→Charge [μm]
P1	503.5	3.5
P2	503.7	3.7
P3	503.8	3.8
P4	503.6	3.6
Average	503.7	3.7
R_a	1.36×10^9 A/Wb	

Tip→Origin = 500 μm Tip→Charge = $|\ell| - 500$
 $N_c = 4$ (4 turns per pole), 4-fold symmetric quadrupole, full-mesh 3D fit ($N = 17772$ FEM nodes total).